

STARTING RECOMMENDATIONS FOR FREE DISPENSE.

These are recommended free dispense initial parameters to help engineers develop their system. As every system is different, our team will work with you on parameter optimization.

Dispense type	Single pipetting																			
Reagent	Water					Ethanol									Serum					
DiTi size	50µL DiTi	200µL DiTi	1000µL DiTi			50µL DiTi			200µL DiTi			1000µL DiTi			200µL DiTi			1000µL DiTi		
Volume (µL)	15-50	25-200	10-20	20-200	200-1000	1-3	3-10	10-50	3-10	10-45	45-200	10-45	45-200	200-1000	5-10	10-45	45-200	10-50	50-200	200-1000
Aspirate speed (µL/s)	50	100	20	100	180	20	20	30	20	50	100	70	150	150	50	100	100	50	100	150
Leading airgap (µL)	15	10	10	20	20	10	10	10	10	10	10	5	5	10	15	20	20	20	20	20
Speed (µL/s)	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70
Trailing airgap (µL)	1.8	5	5	8	10	2	2	2	3	5	5	5	7	10	5	6	6	10	15	15
Speed (µL/s)	20	70	20	70	70	20	20	20	50	50	70	50	70	70	50	70	70	70	70	70
Dispense speed (µL/s)	600	200	400	400	350	300	250	250	300	250	200	400	300	300	400	300	200	500	350	320
Breakoff speed (µL/s)	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50

Dispense type	Single pipetting						
Reagent	50% Glycerol	87% Glycerol	DMSO				
DiTi size	200µL DiTi	200µL DiTi	50µL DiTi	200µL DiTi	1000µL DiTi		
Volume (µL)	20-200	0.5-200	15-50	25-200	10-20	20-200	200-1000
Aspirate speed (µL/s)	10	5	10	100	100	100	100
Leading airgap (µL)	2	2	20	10	10	25	45
Speed (µL/s)	20	20	50	40	20	20	20
Trailing airgap (µL)	1	0	10	10	10	25	20
Speed (µL/s)	20	20	5	20	20	20	20
Dispense speed (µL/s)	10	5	110	250	375	375	500
Breakoff speed (µL/s)	30	30	50	50	50	50	50

Dispense type	Multipipetting	
Reagent	Water	
DiTi size	200µL DiTi	1000µL DiTi
Volume (per aliquot)	10µL	100µL
Aspirate speed (µL/s)	30	150
Delay (ms)	12500	12500
Excess volume (µL)	40	90
Conditioning volume (µL)	30	100
Dispense speed (µL/s)	200	300
Breakoff speed (µL/s)	150	100
Slope	10	20

General rule of thumb to follow for multipipetting:

- 1. Best CVs between 4 and 12 aliquots
- 2. No LAG (leading air gap)
- 3. No TAG (trailing air gap) except for 1000 µL DiTis = TAG after each dispense
- 4. Increase breakoff speed for multipipetting
- 5. CV of an optimized multipipetting is about +1% compared to single pipetting

Definitions:
Aspirate speed: Top speed at which fluid is aspirated.
Leading airgap: Volume of air aspirated before the fluid. This doubles as an air blowout to aid with dispensing.
Speed for leading airgap: The speed at which the leading air gap is aspirated.
Trailing airgap: Volume of air aspirated after the fluid is in the tip.
Speed for trailing airgap: The speed at which trailing air gap is aspirated.
Dispense speed: Top speed at which the fluid is dispensed.
Breakoff speed: It is the cutoff speed at which the dispense is ended.